

HCIVR: An AI-Personalized Virtual Reality Meditation Environment for Stress Reduction

Final Project Report (CS 449/549 – Human Computer Interaction)

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Public repository and materials:

<https://github.com/kadir-sen/SU-CS449-549-Final-Project-HCIVR.git> (demo video, slides, report sources, and evaluation data)

1 Introduction

Digital stress and burnout have become a routine part of everyday life, especially for university students who navigate constant deadlines, dense schedules, and continuous online exposure. Although meditation and wellness applications are now common, many of them are built around static content and assume that users can easily “switch into” a calm mental state by following instructions. In practice, this assumption often fails precisely when users need support the most: during moments of fatigue, anxiety, or cognitive overload.

HCIVR was developed as a response to this gap. The project investigates whether an immersive VR environment—combined with adaptive, AI-supported guidance—can help users relax with less effort. Rather than treating relaxation as a skill users must actively perform, HCIVR treats it as an experience that can be scaffolded through interaction design: clear feedback, minimal decision-making, and a coherent end-to-end user journey.

From an HCI perspective, the core contribution of HCIVR is not a technical pipeline by itself, but a user-centered design and evaluation of an adaptive interface in a sensitive domain. The project therefore emphasizes the Four Pillars of Design framework: (1) clearly defined user expectations and context, (2) theory-grounded process and documentation, (3) implementation as a realized interface artifact, and (4) evaluation with real user interaction.

2 Pillar 1: User-Interface Requirements (User Expectations & Context)

2.1 Target users and domain context

The primary target users are Sabancı University students experiencing moderate everyday stress. The system was intentionally designed to be approachable for users who are new to VR. In our study, the majority of participants (9 out of 14) were first-time VR users, which made learnability, clarity, and perceived safety central requirements rather than optional qualities.

The domain context is digital wellbeing and stress reduction. This context imposes two constraints that shaped our requirements early: first, stressed users have limited patience for confusing interfaces; second, users are particularly sensitive to uncertainty and loss of control during emotionally loaded interactions. A relaxing system must therefore avoid ambiguous states and must communicate what is happening in a way that feels reassuring rather than intrusive.

2.2 User need and problem framing

We observed a recurring mismatch in many meditation tools: users are expected to choose settings, interpret instructions, and sustain focus, even though stress often reduces the very capacity required for these actions. In other words, the interface asks users to do “work” in order to relax.

HCIVR addresses this by aiming for a low-effort interaction model: users should be able to start quickly, provide a minimal emotional input, and then be guided through a coherent session that provides closure. The interface is not designed as a feature showcase, but as a journey with clear stages and predictable transitions.

2.3 Scenario and user journey

A typical scenario is a student finishing an intense day and wanting a short reset before continuing tasks. In this setting, the system must (i) reduce setup complexity, (ii) communicate that the system is responding correctly, and (iii) avoid requiring repeated manual decisions.

The intended user journey is therefore: (1) start session, (2) express current mood or need, (3) receive adaptive guidance and environment behavior, and (4) end with an explicit confirmation that the session concluded successfully. This journey guided our design choices more strongly than any single visual aesthetic decision.

3 Pillar 2: Guidelines, Documents & Design Process

3.1 Theoretical grounding

We grounded our design decisions in core HCI principles that are especially relevant in high-uncertainty or emotionally sensitive contexts. Norman’s Gulfs of Execution and Evaluation provided a practical lens: users should not struggle to map intentions (“I want to calm down”) to actions, and they should not struggle to interpret the system state once they act. Related to this, visibility of system status was treated as a primary design rule, because silent or ambiguous processing states can easily trigger anxiety.

We also treated cognitive load reduction as a design goal. Rather than presenting multiple choices and configurations, we aimed to keep the interface “light” and to move complexity away from the user. Finally, because HCIVR includes adaptive behavior, we considered trust calibration: adaptations should be understandable and should not feel like unexplained automation.

3.2 Process and response to milestone feedback

Our process followed an iterative prototyping approach. Early stages focused on feasibility and building a working VR prototype (Milestone 1). Instructor feedback after Milestone 1 was clear: the submission communicated system architecture but did not provide direct access to an interaction artifact, making HCI-level evaluation difficult. This feedback strongly shaped our final phase. We deliberately shifted the center of gravity from “how the system works internally” to “how the user experiences the system end-to-end.”

As a result, the final deliverables were organized around interaction visibility: a recorded demo video that shows the user journey, a structured evaluation protocol, and a report that connects design choices to observed user experience. In practical terms, this meant spending time not only on building features, but also on making the system legible to users at each step of interaction.

3.3 Team workflow and contributions

The project progressed with a role distribution that supported parallel work while keeping the HCI narrative coherent. Lara focused on UI design choices that make interaction states understandable; Ali implemented interaction flows in Unity to ensure the designed journey becomes a functioning experience. Boray supported the backend side to ensure stable communication across components, while Berkay shaped the AI pipelines that enable adaptive guidance. Bahattin developed the evaluation instruments, explicitly combining quantitative and qualitative questions to capture both measurable outcomes

and subjective experience. Finally, Kadir coordinated the user study, recruited participants, ran sessions, and collected data needed for analysis and reporting.

This structure helped us maintain traceability: design intent could be linked to implementation artifacts, and implementation behavior could be linked to user feedback and measured outcomes.

4 Pillar 3: User-Interface Software Tools (Implementation)

4.1 Tools used and why they fit

HCIVR was implemented in Unity and deployed on VR hardware (Meta Quest). Unity was chosen because it supports rapid iteration on interactive VR experiences and enables consistent control over session flow, transitions, and interface overlays. Figma was used during early stages to reason about interface elements and interaction states before implementing them, helping us separate design intent from implementation constraints.

The system supports two modes that were essential for evaluation: *Baseline Mode*, representing a static meditation experience, and *AI-Supported Mode*, representing adaptive guidance based on user input. Designing both modes was not simply a feature decision—it was a methodological requirement to evaluate whether adaptivity meaningfully changes experience.

4.2 Interaction flow as an implemented artifact

The implemented interaction can be described as a small set of stable stages: session start, input capture, processing, guided experience, and session completion. From an HCI standpoint, what mattered most was not the number of functions, but whether users could follow these stages without confusion.

Three interface elements were especially important in making the interaction understandable. First, guidance text was used to confirm that the system is listening during voice input. This reduces hesitation, particularly for first-time users who are unsure whether the device is “doing anything.” Second, a dynamic loading indicator (spinner) was used during processing to avoid a silent waiting period that users might interpret as a crash or failure. Third, a success screen provided a clean closure, making it clear that the session ended intentionally rather than abruptly.

These were not cosmetic improvements. They were introduced precisely because uncertain or invisible system states are known to harm user trust and can undermine relaxation in emotionally sensitive interactions.

5 Pillar 4: Expert Reviews & Usability Testing (Evaluation)

5.1 Study design and participants

We conducted a comparative user study with real participants to evaluate the system as an HCI artifact. A total of 14 Sabancı University members participated, and 9 of them were first-time VR users. This composition was informative: if a relaxation interface fails for first-time users, it fails in realistic adoption settings.

The evaluation was structured around two conditions aligned with our design: Baseline Mode (static) and AI-Supported Mode (adaptive). The study was not intended to prove medical effectiveness. Instead,

it aimed to assess perceived relaxation, comfort, clarity, and user preference—dimensions that are central to interaction quality.

5.2 Measures and why they were selected

We used a mixed-method approach because relaxation and trust are partly measurable and partly experiential. Quantitative questions used 1–10 scales, but each scale was tied to a specific reason:

- **Perceived relaxation:** to capture whether users felt calmer after the session than before.
- **Comfort and ease of interaction:** to evaluate whether the interface feels approachable, especially for first-time VR users.
- **Clarity of system feedback:** to measure whether users understood what the system was doing at key moments (recording, waiting, completing).

Alongside numeric ratings, we asked open-ended qualitative questions because small frictions in emotionally sensitive interfaces often appear as narratives rather than scores. Typical prompts included: “*How did you feel during the session?*” “*Did you feel more relaxed afterward?*” and “*Was there any moment you felt uncertain about what to do or what the system was doing?*” These questions were designed to surface hesitation, confusion, and trust-related concerns.

5.3 Analysis approach and reporting

To analyze the data, we first summarized outcomes using descriptive statistics (means, variability, and distributions) for each condition. Given the limited sample size, inferential statistics were treated as exploratory rather than definitive: where appropriate, we compared conditions using simple hypothesis tests and reported the results transparently. We also examined whether VR familiarity (first-time vs experienced) related to ratings of comfort or clarity, as a lightweight multivariate perspective on how user background may shape experience.

Crucially, qualitative responses were analyzed thematically. We grouped comments around recurring themes such as reassurance, uncertainty, immersion continuity, and perceived guidance quality. These themes were then used to interpret the quantitative trends and to justify final UI refinements.

5.4 Key findings and interpretation

Across participants, preference consistently favored the AI-supported experience. Beyond preference, the more meaningful pattern was how the interface reduced cognitive burden: users reported that the adaptive mode felt more “guided” and less demanding. This effect was especially valuable for first-time VR users, who are more likely to experience uncertainty during interaction.

From a design standpoint, the strongest lesson was that adaptivity alone is not sufficient. Adaptive behavior must be packaged through transparent interaction cues—otherwise, it risks being perceived as unpredictable automation. The guidance text, spinner, and success confirmation were therefore not minor details but essential trust-support mechanisms that made the adaptive system feel safe and understandable.

6 Final Reflection: Design Motto

Designing relaxation without forcing users to think about how to relax.

We chose this motto because it captures the central HCI insight of HCIVR. Traditional meditation experiences often rely on conscious self-management: users must follow steps, maintain focus, and continuously judge whether they are doing it “correctly.” In our study, users repeatedly valued the opposite: an experience that reduces decision-making, provides reassurance through clear feedback, and allows calmness to emerge without effortful control.

This motto also reflects why a purely static or “gold-standard” meditation approach is insufficient in our problem space. When users are stressed, the interface should take responsibility for clarity, pacing, and confidence. In HCIVR, this responsibility was operationalized through interaction design choices that support user trust and cognitive ease, validated through real user interaction and mixed-method evaluation.

7 Conclusion

HCIVR demonstrates how the Four Pillars of Design framework can be applied to an emotionally sensitive interface domain. By grounding requirements in user context, connecting design decisions to HCI theory, implementing interaction flows as concrete artifacts, and evaluating the experience with real users, the project provides evidence that adaptive, AI-supported VR experiences can improve perceived relaxation and interaction comfort—particularly for novice VR users.

Future work could expand the participant pool, test longer-term usage patterns, and further explore how different user backgrounds shape the effectiveness of reassurance and transparency cues. However, even within the scope of this course project, HCIVR highlights a clear HCI principle: in wellbeing interfaces, interaction clarity is not merely usability—it is part of the emotional outcome.